

TrES-3b

R-0507

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_220919 · created 2026-07-20T18:54:17+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459841.7006 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1903 +/- 0.0096
Transit depth (Rp/Rs)^2	3.62 +/- 0.37 [%]
Orbital Inclination (inc)	83.12 +/- 0.6
Airmass coefficient 1 (a1)	1.017 +/- 0.012
Airmass coefficient 2 (a2)	-0.0126 +/- 0.0053
Scatter in the residuals of the lightcurve fit is	1.18 %
Best Comparison Star	#1 - [409, 374]
Optimal Method	PSF photometry
Transit Duration (day)	0.0671 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1903 ± 0.0096 vs 0.16309 (deviation 2.83σ)
Duration fit vs published	0.0671 d vs 0.0567 d (deviation 2.43σ)
Mid-time O–C	3.3 min
Depth SNR	19.8
Residual scatter	1.18 %

Light curve

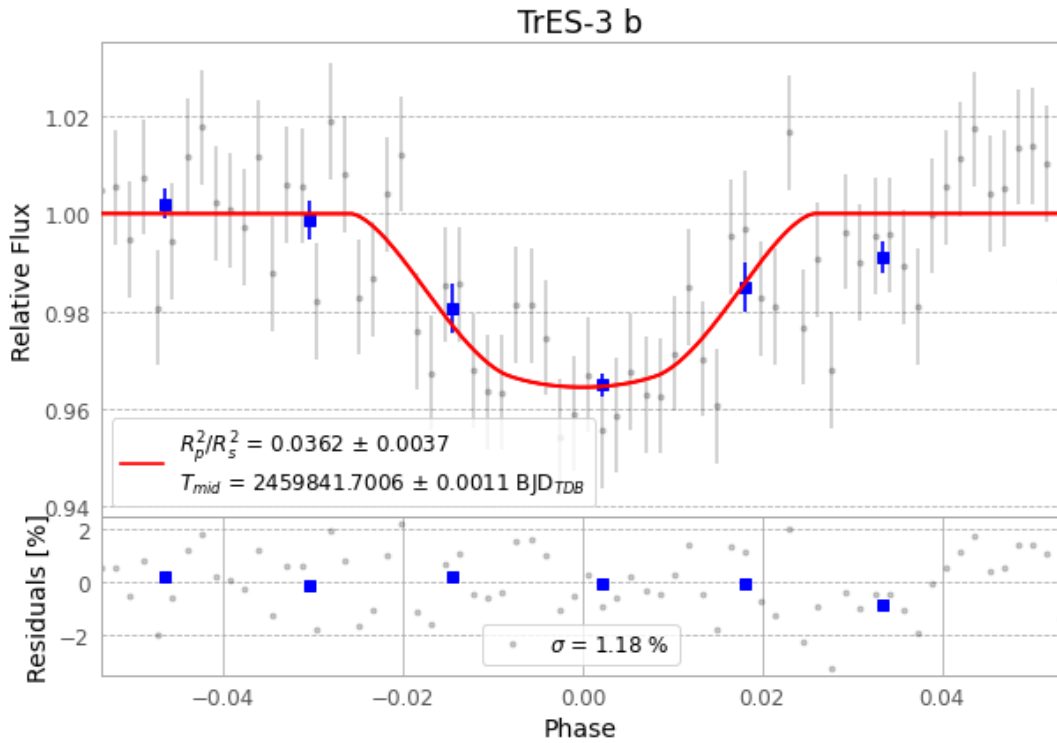


Figure 1 — FinalLightCurve_TrES-3 b_2022-09-18.png (SHA-256 1ceb8425f84cb96b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 216], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 04940c671ae3685f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

68 FITS frames (45 MB), TRES-3220919030615.FITS → TRES-3220919062716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220919.log (e33252f81011...), FinalLightCurve_TrES-3 b_2022-09-18.png (1ceb8425f84c...), FinalLightCurve_TrES-3 b_2022-09-18.csv (64b2c7edcc77...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 18:54Z.